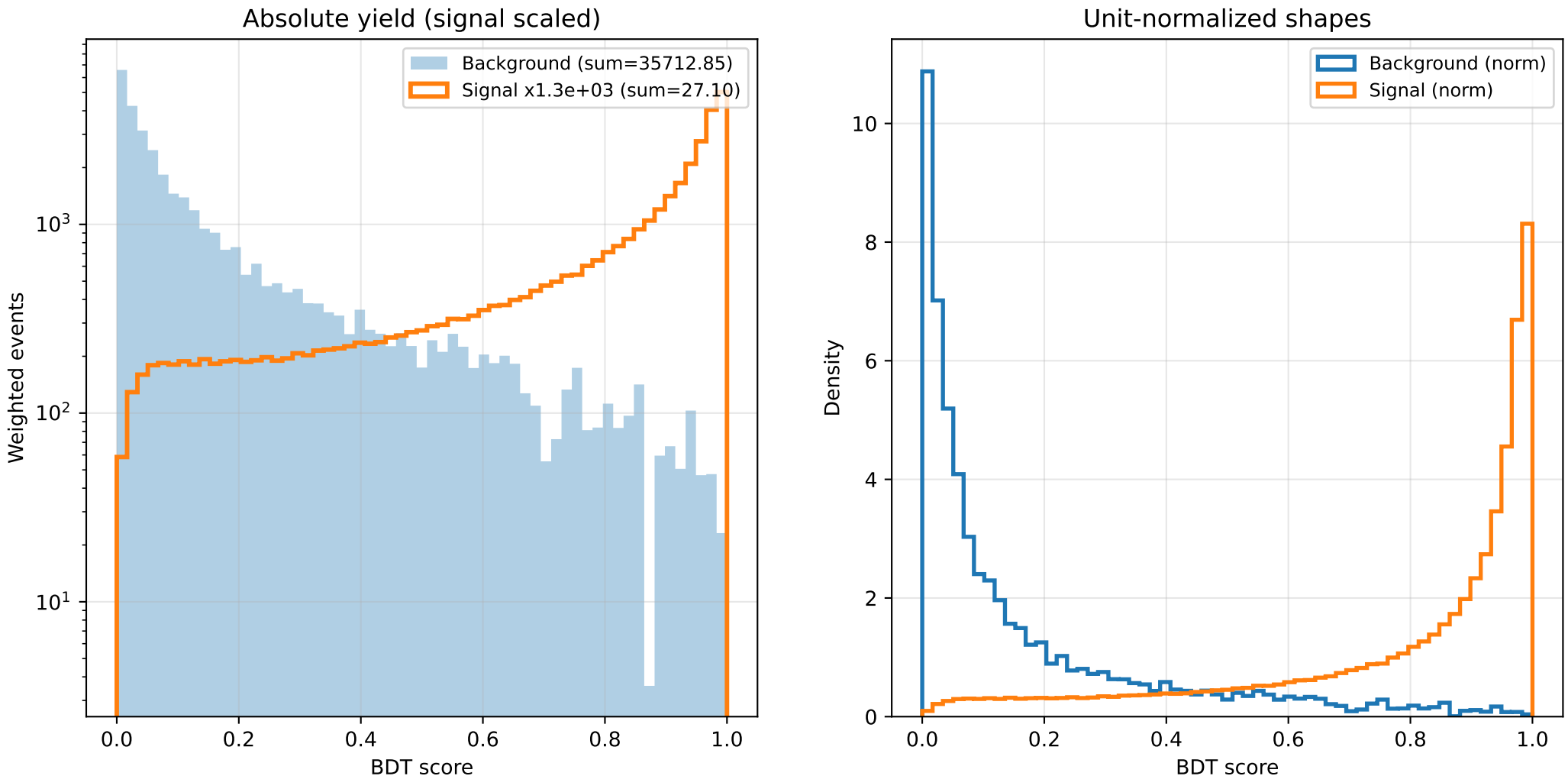
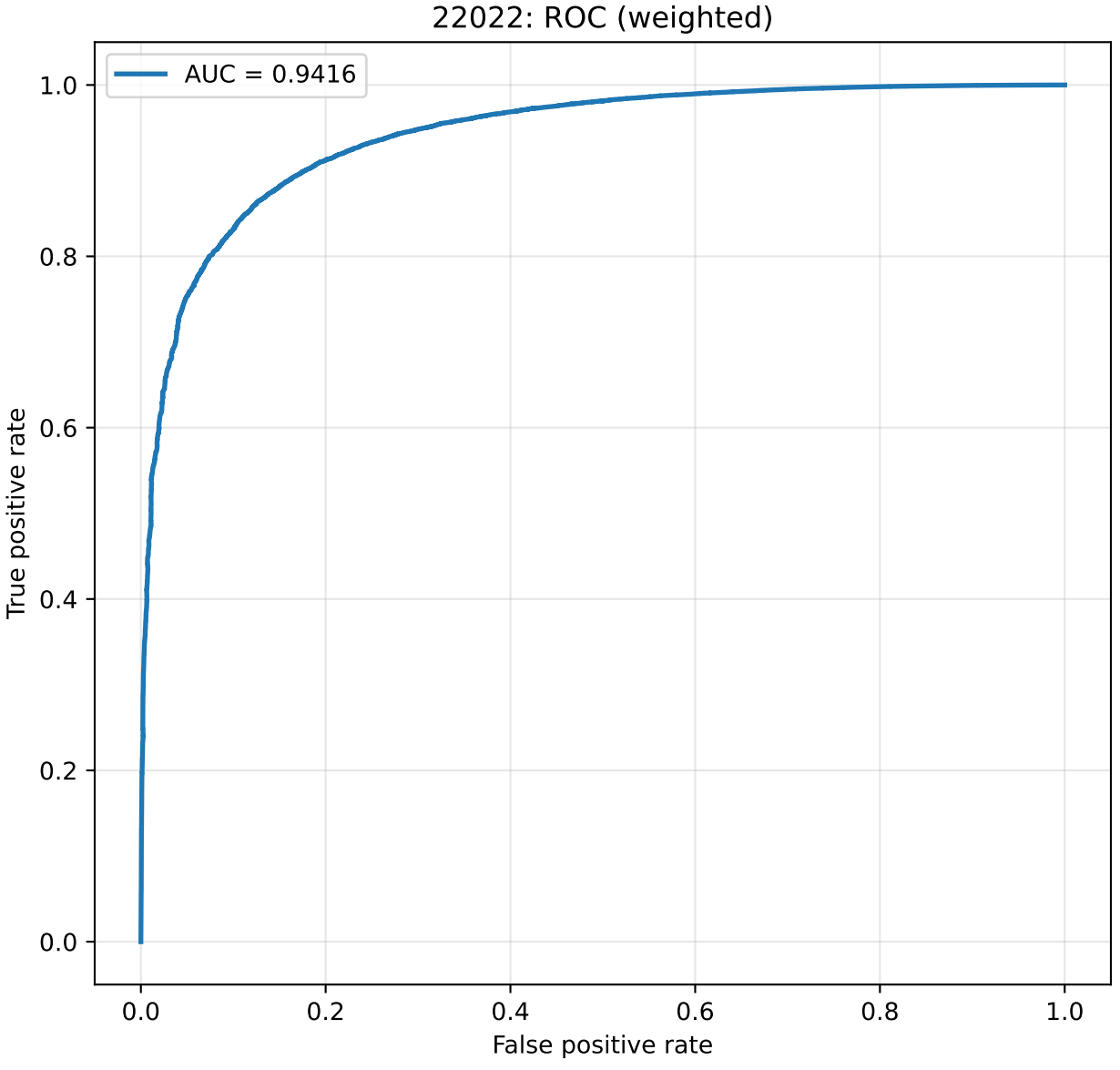
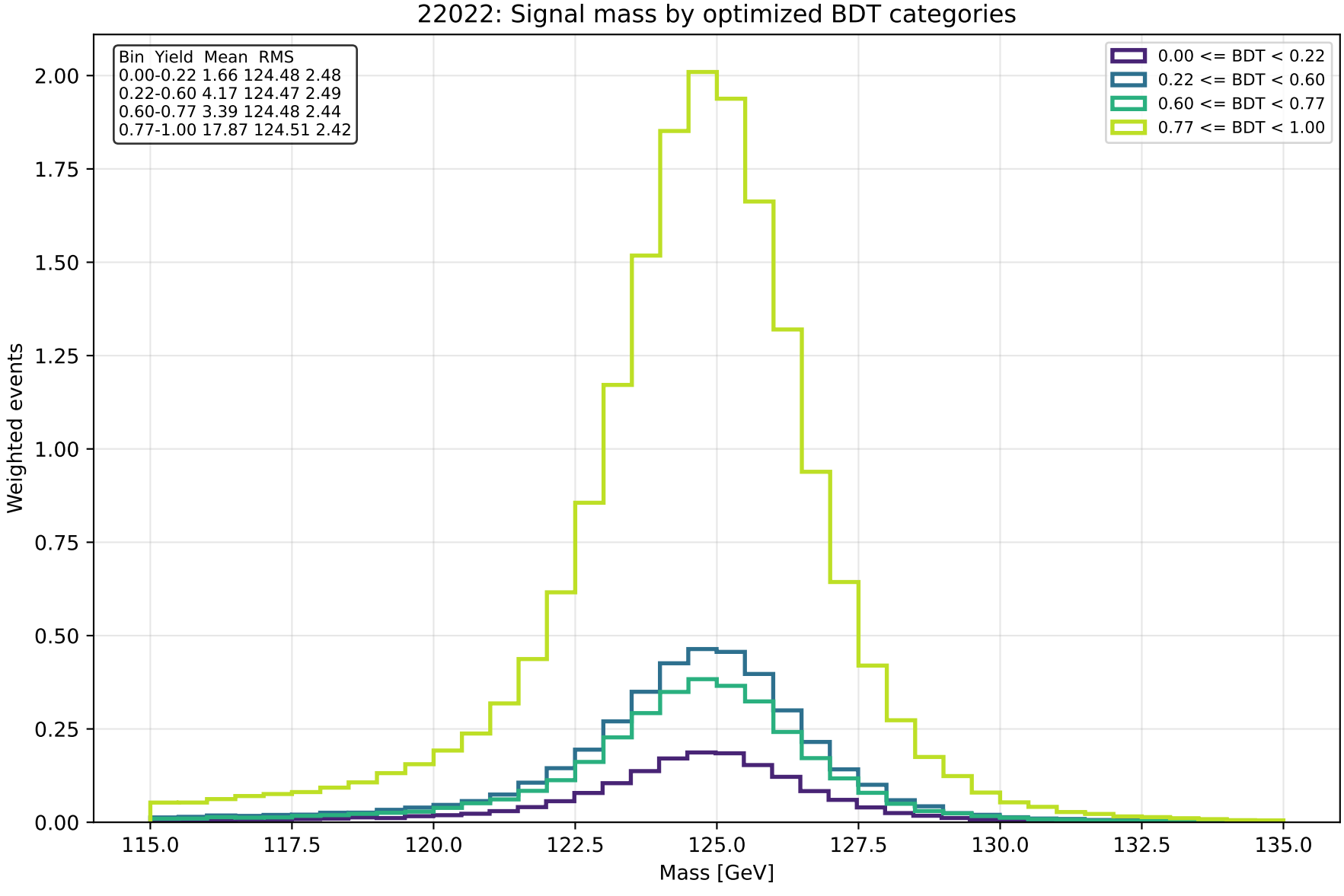
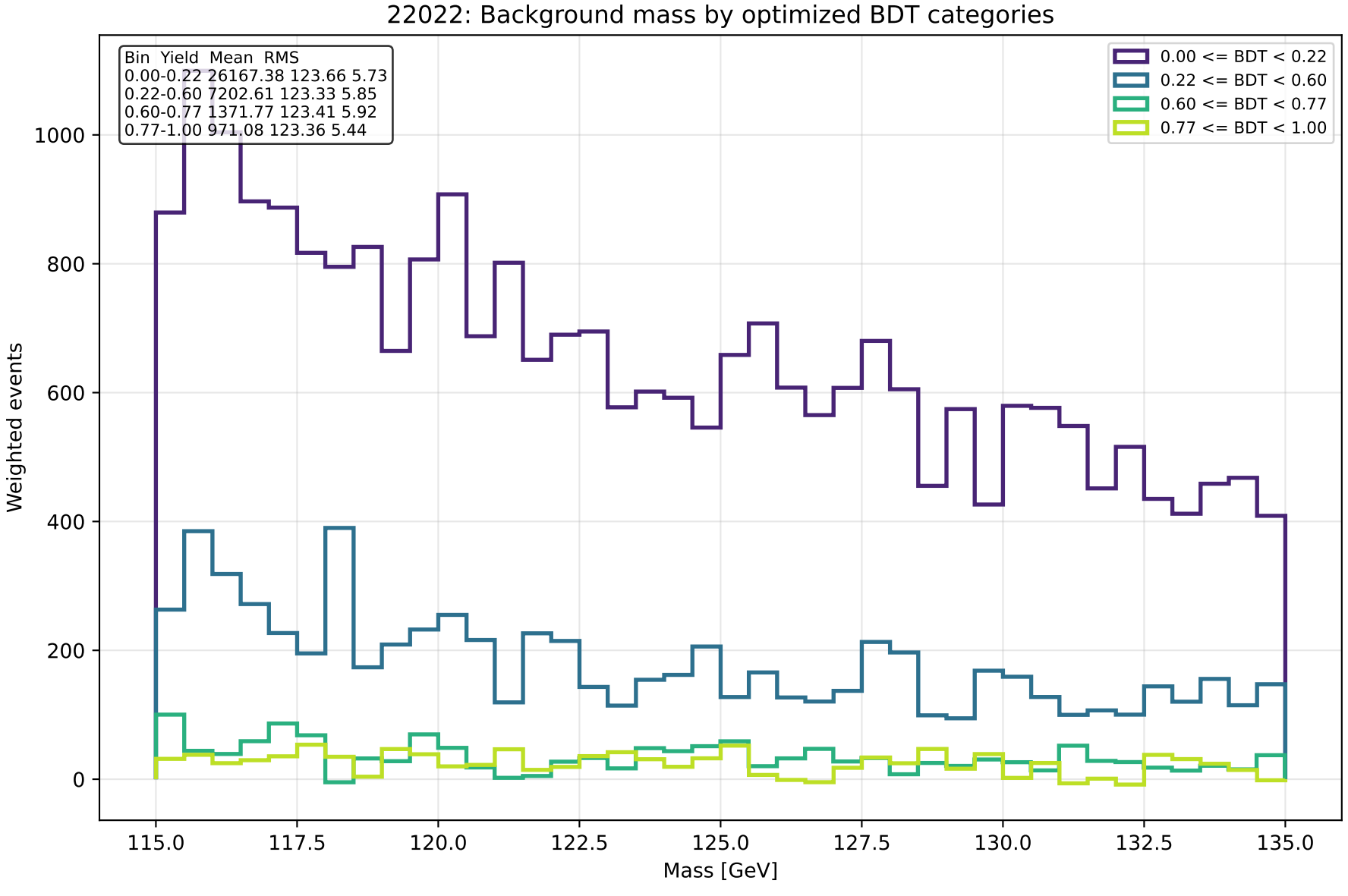

22022: BDT scores in 115-135 GeV
Absolute yield (signal scaled)
Background (sum=35712.85)
Signal x1.3e+03 (sum=27.10)
Weighted events
10^3
10^2
10^1
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

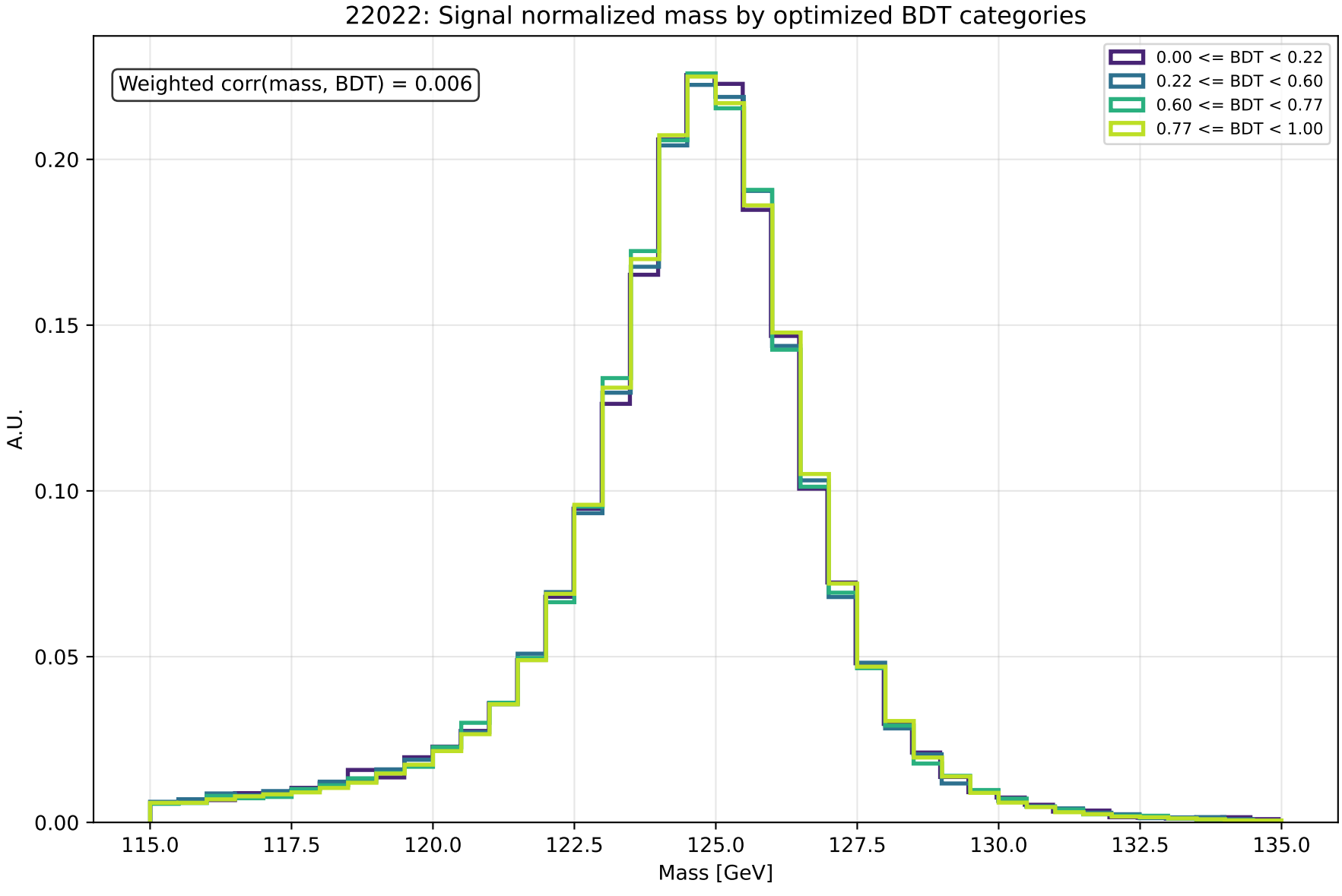

22022: ROC (weighted)
AUC = 0.9416
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

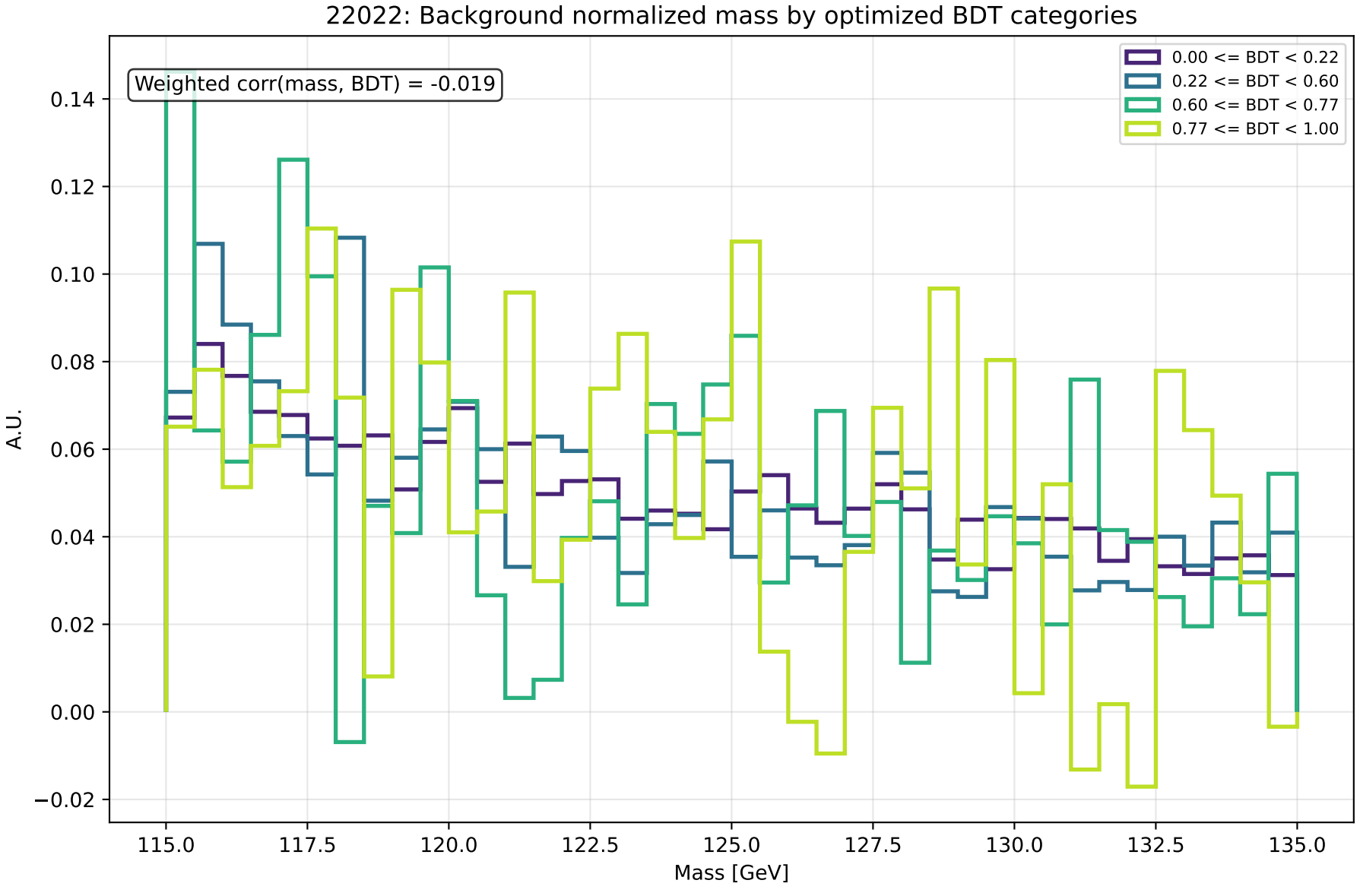
22022: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=2.143, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.219	1.664	26167.383	6.36e-05	0.010
2	0.219-0.599	4.173	7202.613	0.000579	0.049
3	0.599-0.769	3.395	1371.767	0.00247	0.092
4	0.769-1.000	17.867	971.084	0.0184	0.573


22022: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.22 1.66 124.48 2.48
0.22-0.60 4.17 124.47 2.49
0.60-0.77 3.39 124.48 2.44
0.77-1.00 17.87 124.51 2.42
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]


22022: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.22 26167.38 123.66 5.73
0.22-0.60 7202.61 123.33 5.85
0.60-0.77 1371.77 123.41 5.92
0.77-1.00 971.08 123.36 5.44
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
0
200
400
600
800
1000
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]


22022: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.006
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

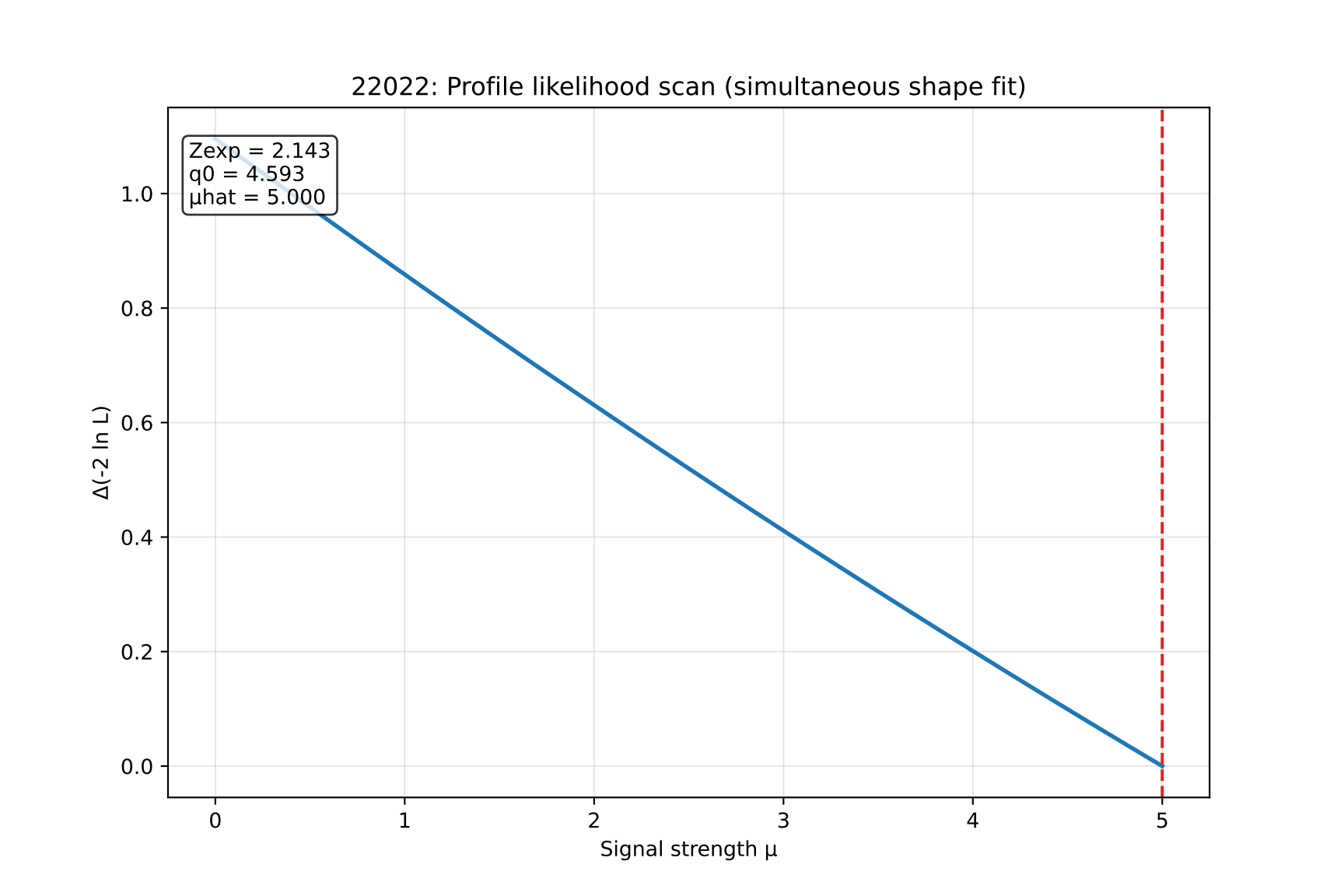

22022: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = -0.019
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

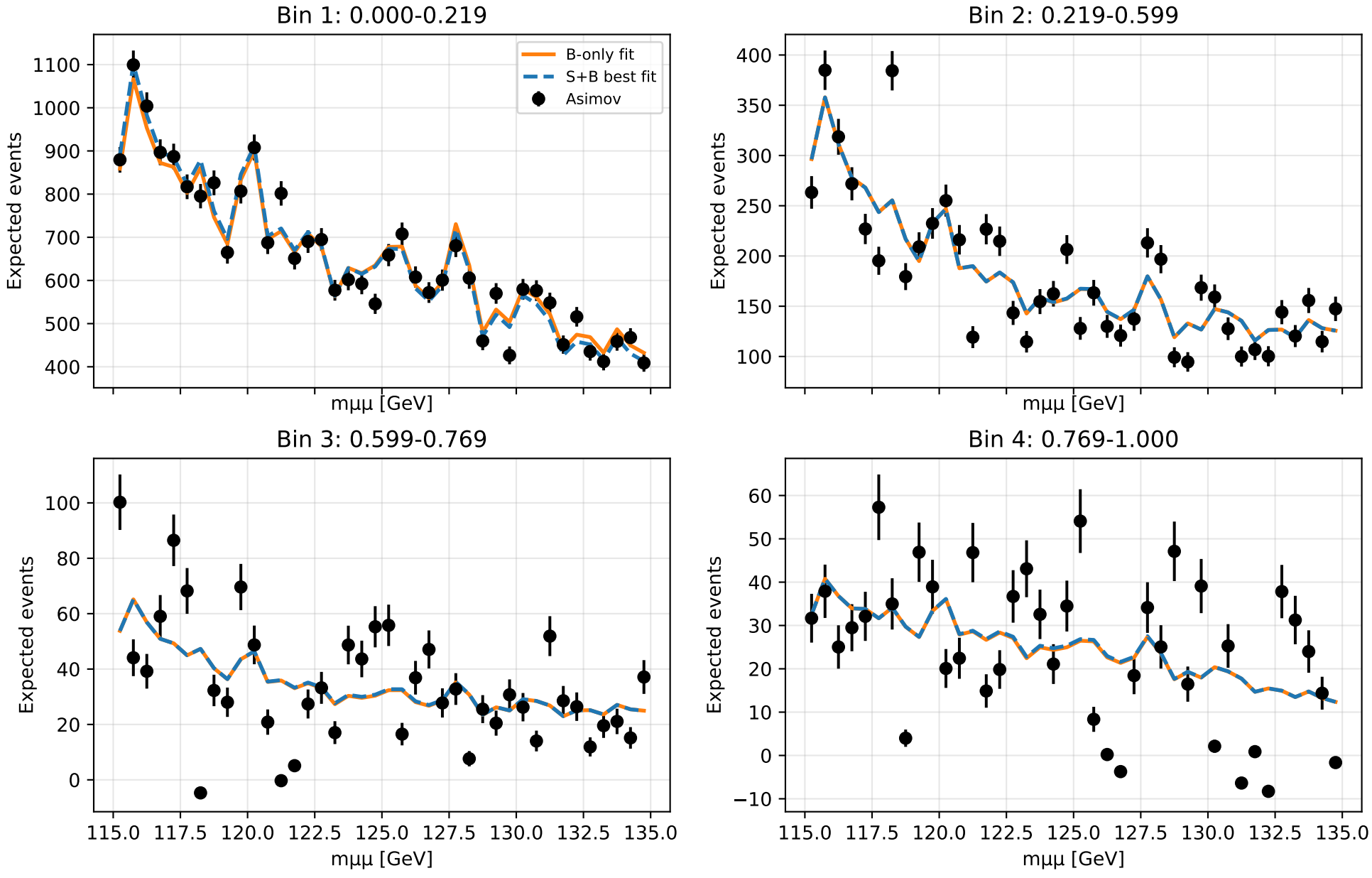
22022: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	2.1432
q0	4.5935
mu_hat	5.0000
N_BDT bins	4
N_mass bins	40
poly order	2
bkg norm unc	0.200
poly unc	0.300
core poly order	2
core from low/med bins	True
DY >=2j unc	0.050
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

22022: Shape-likelihood bin yields (Asimov)

Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.219	1.664	26167.383	26178.679	26172.393
2	0.219-0.599	4.173	7202.613	7205.537	7207.623
3	0.599-0.769	3.395	1371.767	1372.287	1376.777
4	0.769-1.000	17.867	971.084	987.033	991.756


22022: Profile likelihood scan (simultaneous shape fit)
Zexp = 2.143
q0 = 4.593
μhat = 5.000
Δ(-2 ln L)
Signal strength μ
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5

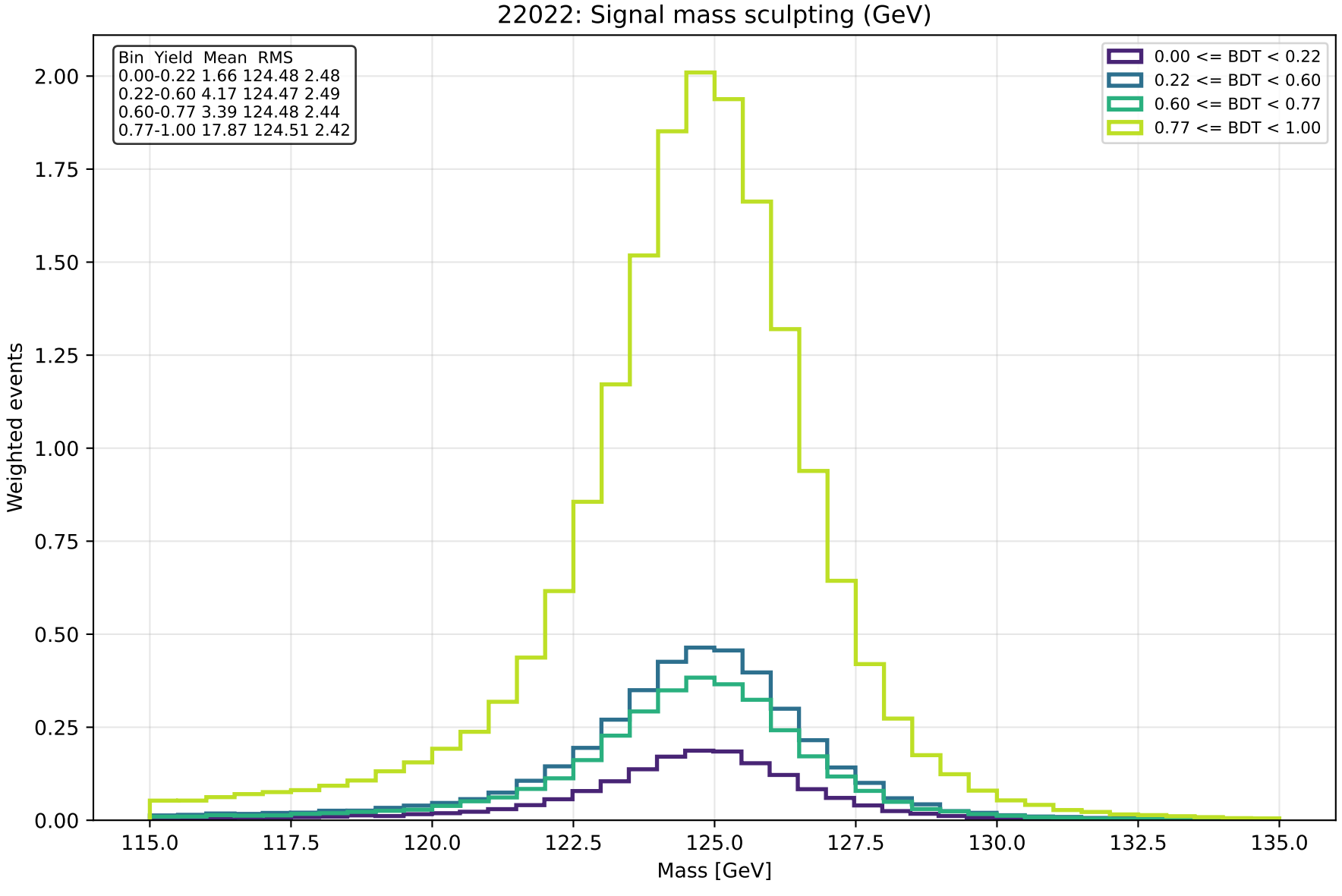

22022: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.219
Bin 2: 0.219-0.599
Bin 3: 0.599-0.769
Bin 4: 0.769-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

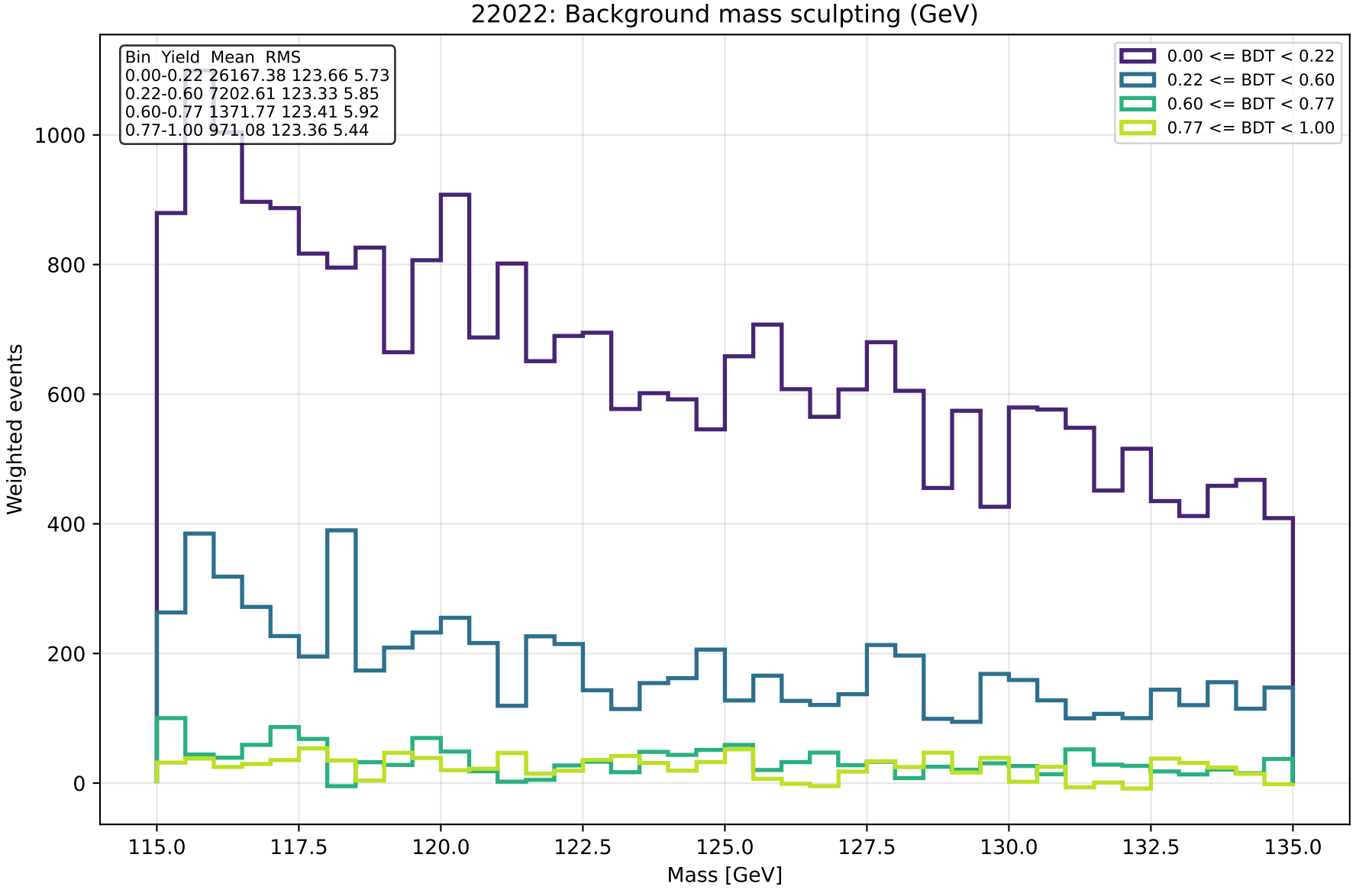
SumExp	LandaUxBernstein	0.0000	0.0000	-1.0000	FAIL
SumExp	BWZxBernstein	2.6733	2.8061	1.6733	FAIL
SumExp	BWZGamma	7.6650	5.0469	6.6650	FAIL
PowerLaw	BWZRedux	2.9133	2.7265	1.9133	FAIL
PowerLaw	SumExp	8.0967	4.2849	7.0967	FAIL
PowerLaw	PowerLaw	2.9667	3.6692	1.9667	FAIL
PowerLaw	FEWZxBernstein	2.7583	3.0847	1.7583	FAIL
PowerLaw	LandaUxBernstein	0.2833	1.1390	-0.7167	FAIL
PowerLaw	BWZxBernstein	2.2933	3.1677	1.2933	FAIL
PowerLaw	BWZGamma	7.5150	5.3995	6.5150	FAIL
FEWZxBernstein	BWZRedux	1.6317	2.1944	0.6317	FAIL
FEWZxBernstein	SumExp	9.3367	4.5751	8.3367	FAIL
FEWZxBernstein	PowerLaw	3.2900	3.6331	2.2900	FAIL
FEWZxBernstein	FEWZxBernstein	2.3433	3.1549	1.3433	FAIL
FEWZxBernstein	LandaUxBernstein	0.0083	0.0449	-0.9917	FAIL
FEWZxBernstein	BWZxBernstein	1.1967	2.2448	0.1967	PASS
FEWZxBernstein	BWZGamma	5.9933	3.7158	4.9933	FAIL
LandaUxBernstein	BWZRedux	395.0150	11.8071	394.0150	FAIL
LandaUxBernstein	SumExp	787.7033	7.1349	786.7033	FAIL
LandaUxBernstein	PowerLaw	493.5883	12.7224	492.5883	FAIL
LandaUxBernstein	FEWZxBernstein	640.3517	4.7124	639.3517	FAIL
LandaUxBernstein	LandaUxBernstein	1.1267	1.4101	0.1267	PASS
LandaUxBernstein	BWZxBernstein	709.5533	10.2671	708.5533	FAIL
LandaUxBernstein	BWZGamma	792.1850	7.2816	791.1850	FAIL
BWZxBernstein	BWZRedux	3.4450	4.3125	2.4450	FAIL
BWZxBernstein	SumExp	6.1683	4.6376	5.1683	FAIL

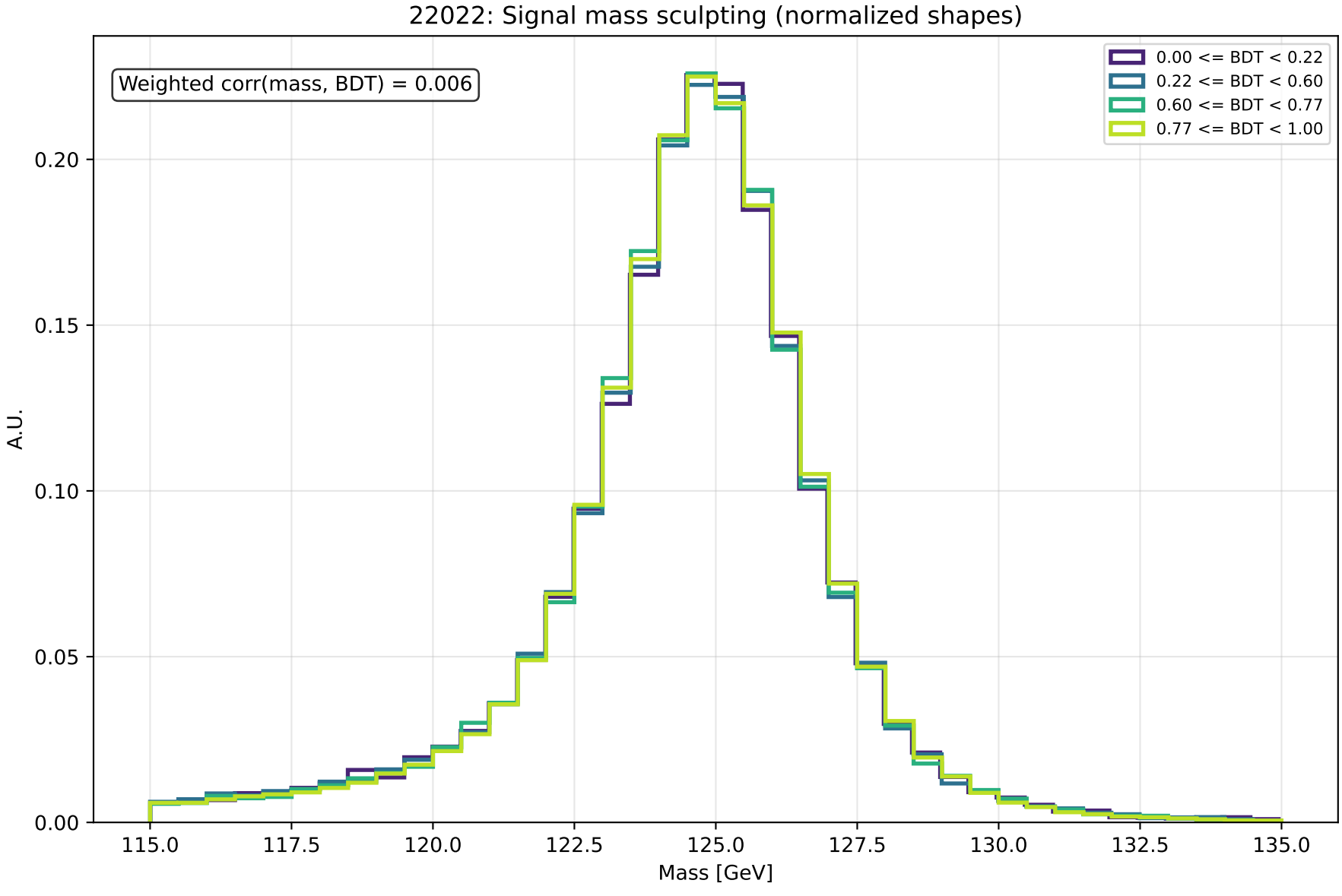
2022 Bias matrix (mu_true = 1.00, threshold = 0.20)

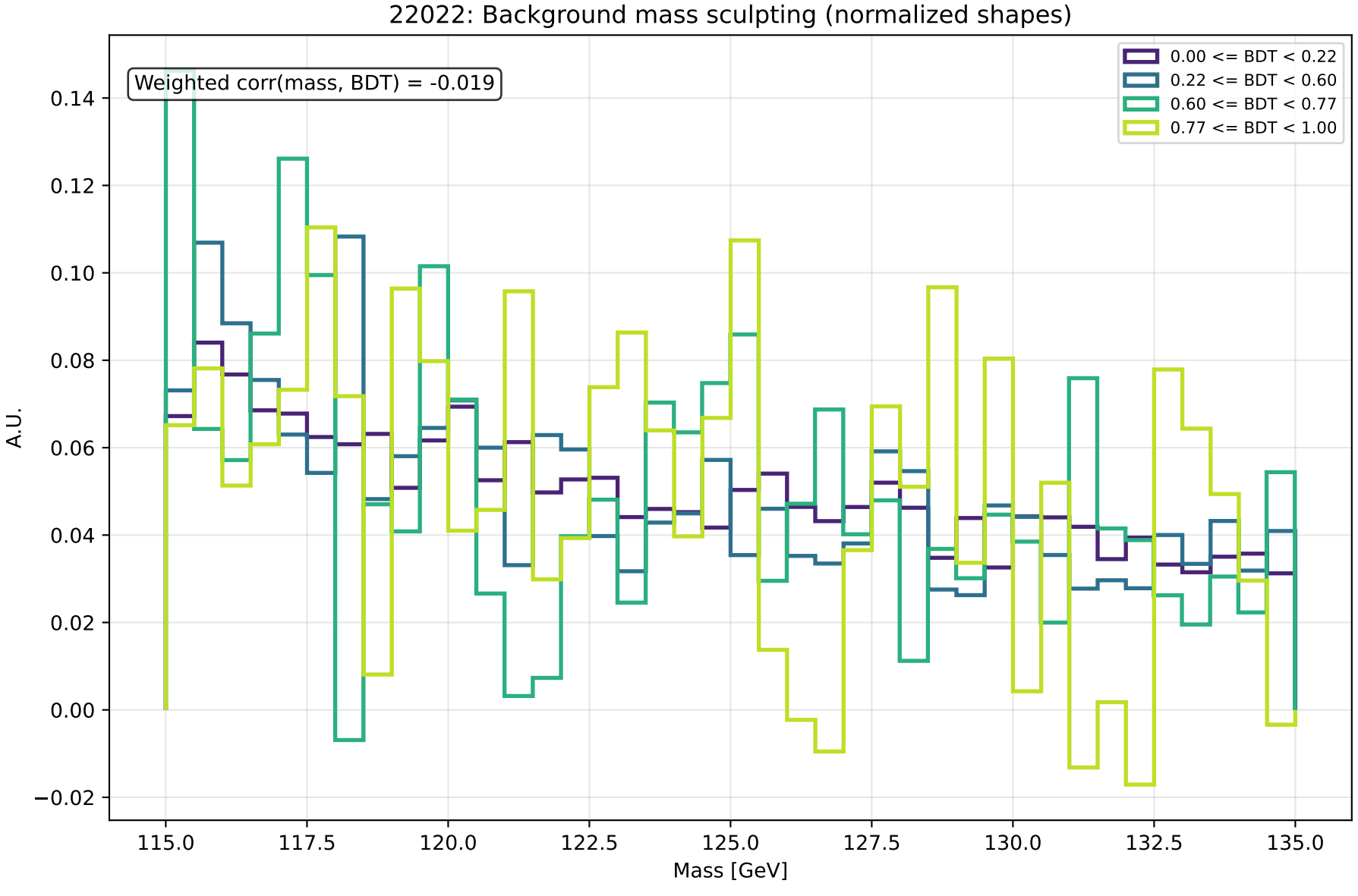
22022: Expected yield per optimized BDT bin (115-135 GeV)
S total=27.10, B total=35712.85

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.219	1.664	6.1%	100.0%	26167.383	0.000759
0.219-0.599	4.173	15.4%	93.9%	7202.613	0.00266
0.599-0.769	3.395	12.5%	78.5%	1371.767	0.00908
0.769-1.000	17.867	65.9%	65.9%	971.084	0.0184

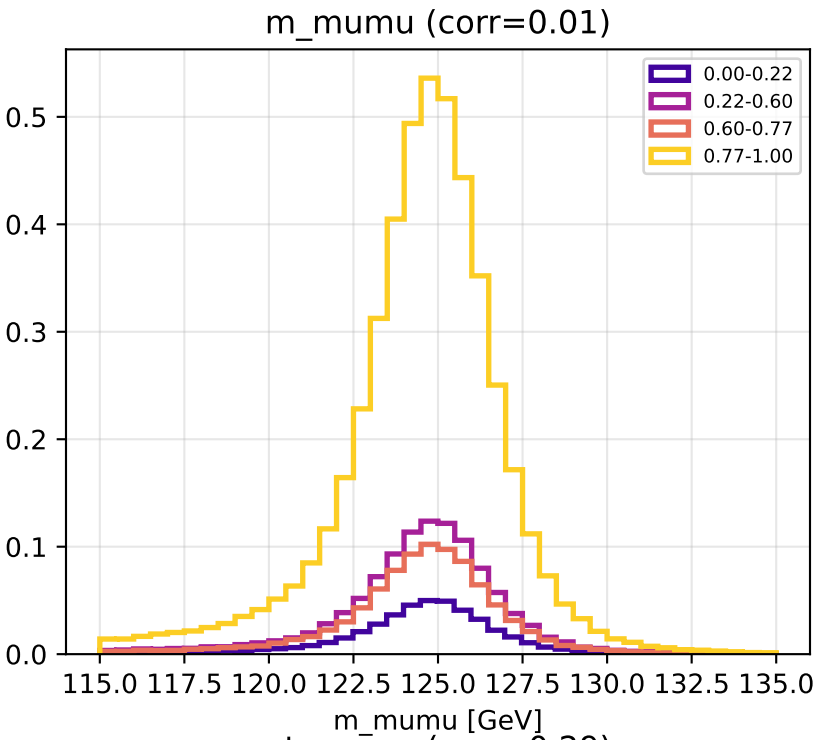

22022: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.22 1.66 124.48 2.48
0.22-0.60 4.17 124.47 2.49
0.60-0.77 3.39 124.48 2.44
0.77-1.00 17.87 124.51 2.42
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

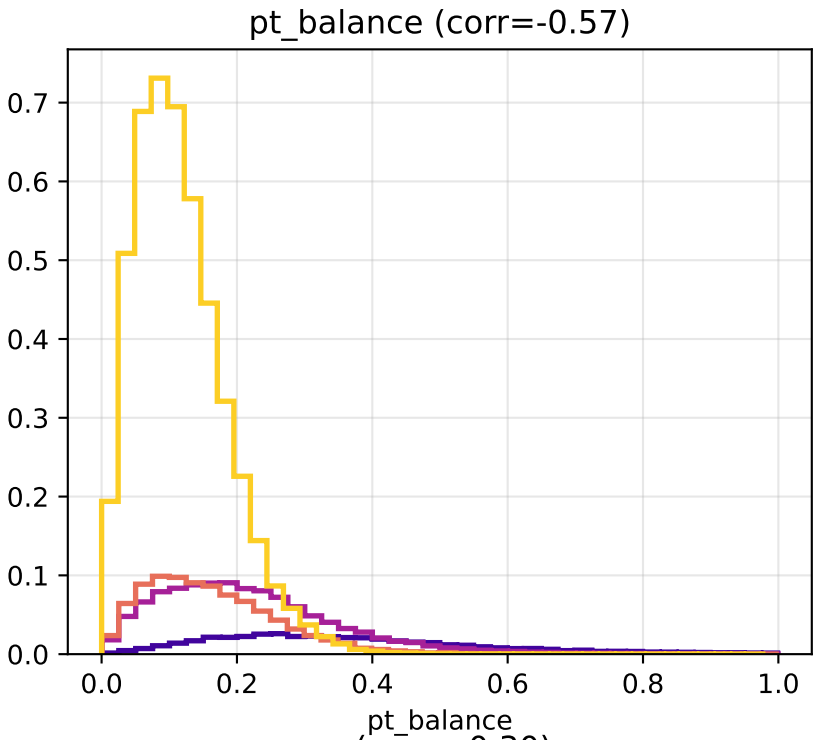

22022: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.22 26167.38 123.66 5.73
0.22-0.60 7202.61 123.33 5.85
0.60-0.77 1371.77 123.41 5.92
0.77-1.00 971.08 123.36 5.44
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
0
200
400
600
800
1000
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

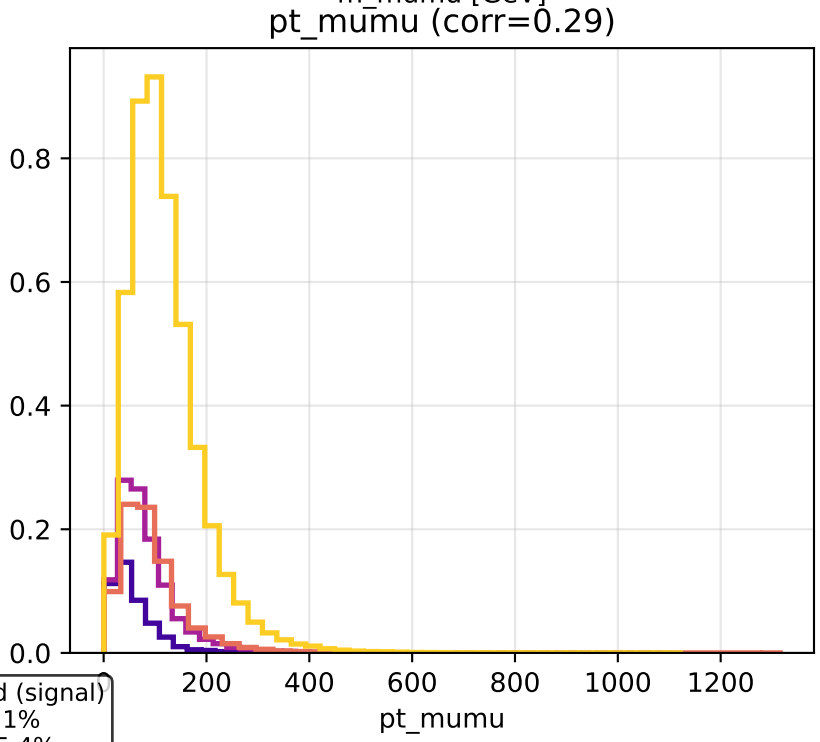

22022: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.006
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

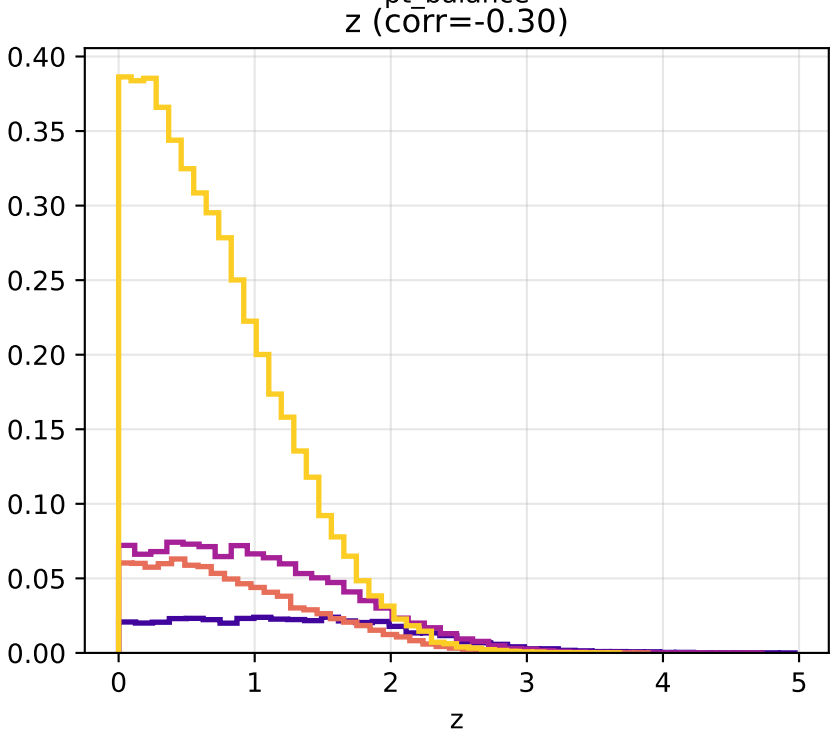

22022: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = -0.019
0.00 <= BDT < 0.22
0.22 <= BDT < 0.60
0.60 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

22022: Signal distributions by BDT bin

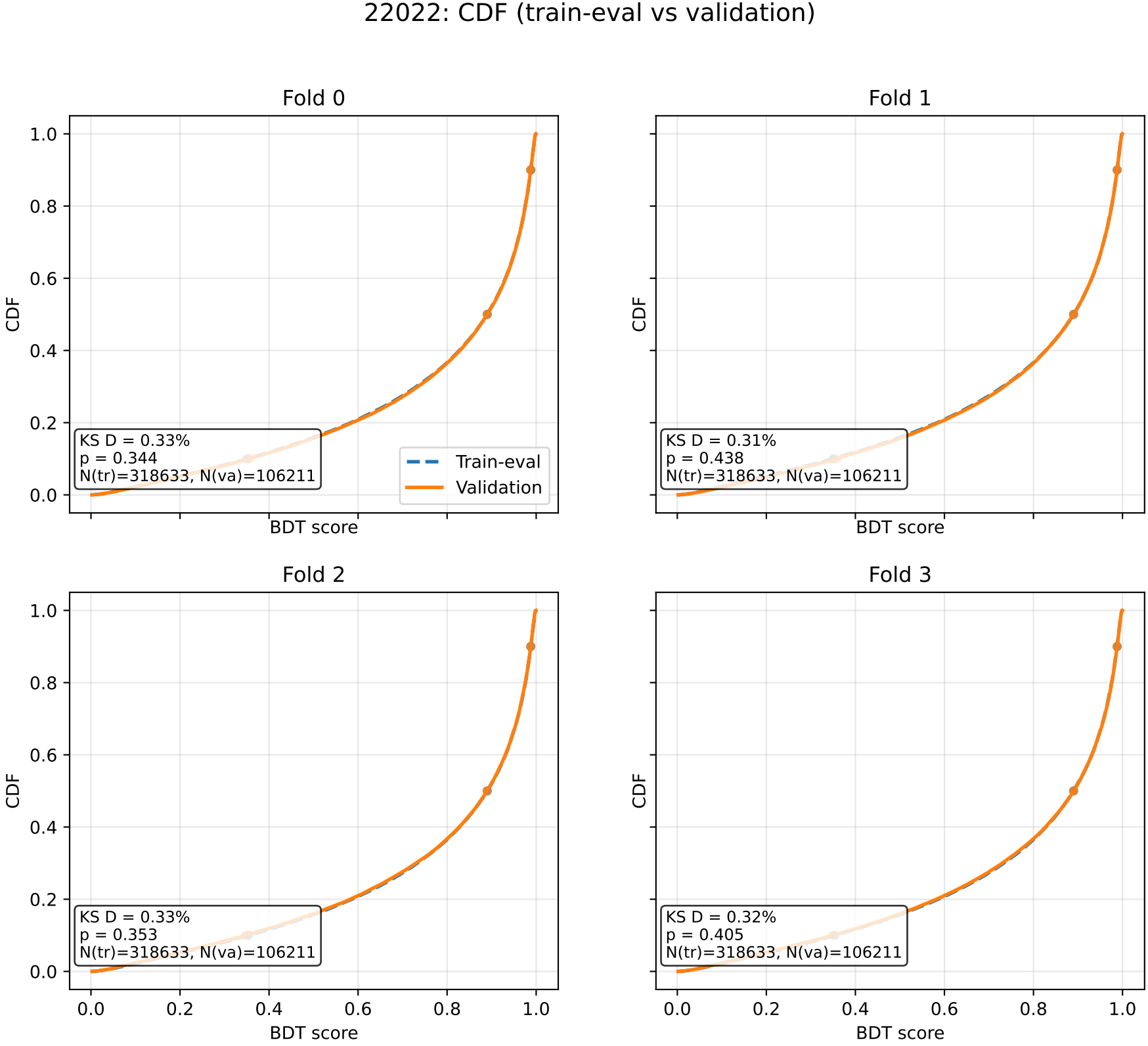








BDT bin yield (signal)
0.00-0.22: 6.1%
0.22-0.60: 15.4%
0.60-0.77: 12.5%
0.77-1.00: 65.9%


22022: CDF (train-eval vs validation)
Fold 0
KS D = 0.33%
p = 0.344
N(tr)=318633, N(va)=106211
Train-eval
Validation
Fold 1
KS D = 0.31%
p = 0.438
N(tr)=318633, N(va)=106211
Fold 2
KS D = 0.33%
p = 0.353
N(tr)=318633, N(va)=106211
Fold 3
KS D = 0.32%
p = 0.405
N(tr)=318633, N(va)=106211
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

22022: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.33%	106211	0.001	0.20%	FALSE
1	0.31%	106211	0.001	0.20%	FALSE
2	0.33%	106211	0.001	0.20%	FALSE
3	0.32%	106211	0.001	0.20%	FALSE

22022: Top-10 feature importance (gain)
22022_featureimportance_DY.txt

Feature	Gain %
pt_balance	22.93%
pt_mumu	9.72%
z	9.36%
dphi_mumu_jj	7.60%
deta_jj	7.10%
njets	6.37%
pt_j2	5.30%
pt_j1	4.85%
jet1_eta	3.36%
mu1_pt_over_m	2.82%